Aarav Aggarwal

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EDUCATION

Singapore American School, Singapore *Aug 2021 - Present*

- Grade 7 - Grade 12, American School Diploma with APs and ATs
- Grade 11 GPA: 4.43 (Sem1); Grade 10 GPA: 4.29; Grade 9 GPA: 4.18

The Shri Ram School Aravali, Gurgaon *Apr 2013 - Jul 2021*

- Till Grade 6, ICSE Indian Board

STANDARDIZED TESTS & APS

- **SAT**: 1570/1600
- **APs**: AP Computer Science (5), AP Microeconomics (5), AP Macroeconomics (5)
  Currently pursuing: AP Calculus BC, AP Physics 1, AP Chemistry, AP African American Studies, AP Psychology

HONORS & AWARDS

School:

- Grade 9 and Grade 10 Scholars List: Awarded to top performing students (GPA of 4.0+)
- Selected for Singapore American School chapter of Rho Kappa (Dec 2025 - present), based on sustained academic excellence in the Social Sciences

International:

- Professional Cricket: Represented Singapore in U16 National team at Asian Cricket Council Tournament in 2025, finished Runner-Up (10 countries)
- Research Publication "Roof of Hope: Evaluating the Thermal and Social Impacts of Cool Roofs in Low-Income Settlements of North India" accepted to 7th International Conference On Counter Measures to Urban Heat Islands, 2026 Conference
- Published Author, Elaine Hobson Essay Competition: 500-word essay, "Ripple Effect" (July 2024)

ENGINEERING & TECHNOLOGY EXPERIENCE

Innovation:

DataPitch | Developer—Data Preprocessing, Visualising, ML Algorithms *Oct 2024 - Present*

- Built an ML model to predict player performance in the Indian Premier League (IPL)—a Global T20 cricket league—using Python models such as K-Means clustering, PCA, and Linear Regression
- Submitted research paper "Batting Archetypes and Team Success: Analyzing the Role Composition of Teams in Indian Premier League Using Machine Learning" to The National High School Journal of Science

Young Technology Scholars | Summer School Participant, Plaksha University, Chandigarh *Jun 2023*

- Highly selective summer program for high school students exploring intersection of technology and real-world applications
- Built a prototype of Remote Surgery Teleoperation Bot with ESP-32, gyroscope, & motors for a hand-controlled robot

Internships:

Mynzo Carbon | Machine Learning Intern — Computer Vision, Sustainability Analytics *Jun - Aug 2025*

- Developed a leaf-species identification pipeline from in-situ photos using MobileNetV2 transfer learning, robust augmentation, and cross-validated evaluation
- Curated a 250-image dataset across 10 species and achieved ~85% accuracy; applied the model to tag field samples, accelerating species-distribution assessment for high-integrity carbon projects

URBAN HEAT STRESS INITIATIVE & RESEARCH

Roof of Hope (Link) | Founder, India *Dec 2024 - Present*

- Collaborated with Mahila Housing Trust and Indorama Ventures for Clean Energy, Plaksha University to pilot and test solutions for building resilience to urban heat stress for low-income communities in Delhi
- Installed cool roofs in 100 low-income homes in Bakkarwala, in Delhi to provide relief in extreme summer months, and to collect & analyse data for research and policy planning on urban heat stress relief
- Findings reveal a significant reduction of ~20 °C in over-deck surface temperatures, 4–11 °C in under-deck ceiling temperatures, and 1–3 °C in indoor wall temperatures. Structured surveys and qualitative interviews confirm substantial improvements (reported by 95% of surveyed households) in thermal comfort, sleep quality, daily activities, and self-reported health

Cool Play | Founder, India *May - Aug 2025*

- Partnered with the MIMA (Mera India Mera Adhikar) Foundation to run a community driven fundraising campaign raising $1,000+ to support young athletes from low-income backgrounds in Delhi
- Distributed 60+ cooling jackets to young athletes, resulting reduced incidents of heat exhaustion and safe training
- Inspired local schools and sports clubs to explore similar cooling jackets creating a ripple effect beyond direct beneficiaries

TEACHING & MENTORING

Plaksha University | Teaching Assistant, Chandigarh, India *Jun 2024*

- Invited to be a Teaching Assistant at Young Tech Scholars program
- Assisted 80+ students on a Teleoperation project through live lessons, instructional videos and hands-on demonstrations

Read to Lead | Vice President, Singapore American School *Aug 2024 - Present*

- Partnered with the Singapore National Library Board to launch monthly literacy workshops, reaching 100+ young readers through interactive sessions
- Co-designed curriculum and initiated a book drive to provide underprivileged children with access to 500+ books

EXTRACURRICULARS

Singapore Cricket Association | Singapore U16 National Team & U19 High-Performance Squad *Jan 2018 – Present*

- Selected to represent Singapore U16 National Cricket Team at the Asian Cricket Council (ACC) U16 Championship; achieved Runner-Up finish in 10 country tournament
- Played competitively for 8 years, train 6 days a week with National High-Performance Squad, Div 1 in SCA Men's League

Cricket Club | Officer, Singapore American School. *May 2024 - Present*

- Led organization of 5+ inter-school events and 2 international tours (Philippines, Malaysia) to grow cricket participation and cross-border engagement
- Raised funds and implemented structured training upgrades, including permanent nets and a development program benefiting 40+ student athletes

Stumps and Stories | Head of Logistics, Singapore American School *Oct 2024 - Present*

- Co-led planning and execution of a cricket tournament honouring over 80 migrant workers and staff, fostering community appreciation and inclusivity through sport and storytelling
- Designed awareness campaigns and contributed to a $5,000 fundraising effort to support event operations, refreshments, and commemorative kits

SKILLS

Technical: Proficient in Python, Java, Familiar with Arduino IDE
Languages: English (Native), Hindi (Native), Spanish (Conversational)
Interests: Deep Sea Diving (PADI Certified Diver), Cricket, Sports, International Politics, Debating